recommendations. There is good evidence for the safety of intensive LDL-cholesterol lowering, and statin-based therapy combined with a fire-and-forget strategy is low cost. This approach is consistent with a more recent recommendation for primary prevention in CKD by the American College of Cardiology/American Heart Association (which recommended 10-year thresholds of >7.5%).[674]

Practice Point 3.15.1.4: Consider prescribing proprotein convertase subtilisin/kexin type 9 (PCSK-9) inhibitors to people with CKD who have an indication for their use.

Proprotein convertase subtilisin/kexin type 9 (PCSK-9) inhibitors have been shown to safely reduce ASCVD risk when added to maximal tolerated statin-based regimens in people at high coronary risk.[675,676] Subgroup analyses suggest that their safety profile and their biochemical and clinical efficacy are similar when participants with CKD and without CKD are compared. These trials recruited down to an eGFR of 20 ml/min per 1.73 m^2.[677,678] Current examples of recommendations for the use of PCSK-9 inhibitors from the cardiology community (and licensed indications) include as an adjunct to diet and maximally tolerated statin therapy for the treatment of adults with heterozygous familial hypercholesterolemia or for people with clinical ASCVD who require additional lowering of LDL cholesterol.[679,680]

Dietary approaches.

Practice Point 3.15.1.5: Consider a plant-based "Mediterranean-style" diet in addition to lipid-modifying therapy to reduce cardiovascular risk.

Diet and lipids have been comprehensively reviewed by other clinical practice guidelines.[679,681] In that work, the Work Groups highlighted that in general populations, observational studies have associated plant-based diets that include higher consumption of fruit, vegetables, nuts, legumes, fish, olive oil, yogurt, and whole grains, with lower risk of CVD. Diets associated with higher risk are those including high consumption of red and processed meats, refined carbohydrates, and salt. Vegetable sources of fats and polyunsaturated fatty acids (e.g., in nuts, seeds, avocado, and olive oil) are also associated with a lower risk compared with animal fats, including dairy fat.[679] A Mediterranean-style diet has an emphasis on extra virgin olive oil and is high in unsaturated fat. RCTs have shown that such diets have important effects on cardiovascular risk in the long term despite only small effects on traditional markers of metabolic syndrome profile.[682–685] In the large Prevención con Dieta Mediterránea (PREDIMED) primary prevention trial of 7447 adults, the Mediterranean diet rich in extra virgin olive oil reduced the risk of major cardiovascular events by 31% (HR: 0.69; 95% CI: 0.53–0.91). The Coronary Diet Intervention With Olive Oil and Cardiovascular Prevention (CORDIOPREV) trial found that allocation to a Mediterranean diet rich in extra virgin olive oil reduced the risk of the composite of MACE by approximately 22%–25%.[684] There is no large-scale CKD-specific trial comparing these dietary interventions.

3.15.2 Use of antiplatelet therapy

Recommendation 3.15.2.1: We recommend oral low-dose aspirin for prevention of recurrent ischemic cardiovascular disease events (i.e., secondary prevention) in people with CKD and established ischemic cardiovascular disease *(1C)*.

This recommendation places high value on the importance of reducing recurrence of myocardial infarction, ischemic strokes, or peripheral arterial disease complications in people with CKD and established ischemic CVD due to the mortality and disability associated with such complications. In secondary prevention, trials have clearly shown the absolute benefits of low-dose aspirin substantially exceed the potential for bleeding complications, creating certainty about net benefits when treating this population. In people with CKD without prior ischemic CVD, the balance of benefits and risks are uncertain and may be counterbalanced—large RCTs are ongoing.

Key information

Balance of benefits and harm. Based on a number of large RCTs in populations that are likely to be largely free from CKD, lifelong use of low-dose aspirin (75–100 mg) for the prevention of recurrence of complications of ischemic CVD is strongly recommended among people with known CVD (a therapeutic approach referred to as secondary prevention). Conversely, it is not possible to provide definitive recommendations on when to use aspirin to prevent a first ischemic cardiovascular event (i.e., primary prevention) in people at high risk, and a research recommendation is provided. This is due to uncertainty of the net absolute value of such an approach, as any reduction in the risk of atherosclerotic cardiovascular events needs to be weighed against the risk of major bleeding. It is important to consider CKD-specific data in the totality of the evidence.

Key evidence from general populations is derived from a 2009 meta-analysis by the Anti-thrombotic Treatment Trialists' collaboration. The analyses included data on long-term aspirin use versus control care in 16 secondary prevention trials (approximately 17,000 people at high average risk, approximately 43,000 person-years, 3306 serious vascular events [defined as myocardial infarction, stroke, or cardiovascular death]), and 6 primary prevention trials (approximately 95,000 participants at low average risk, approximately 660,000 person-years, 3554 serious vascular events).[685a] In the secondary prevention trials, allocation to aspirin reduced the risk of both ischemic stroke and myocardial infarction by about one-fifth, such that an overall RR reduction for any serious vascular event was 19% compared with controls (RR: 0.81; 95% CI: 0.75–0.87). This equated to a 1.49%

lower absolute risk of serious vascular events per year compared with an estimated absolute risk of any major bleeding, which was an order of magnitude smaller at 0.03% per year. Note that this hazard of major bleeding was extrapolated from the primary prevention trials as stroke causes and extracranial bleeds were generally not well recorded in the relatively older secondary prevention trials (Figure 38[685a]).

Some people with CKD have been included in antiplatelet therapy trials. A recent Cochrane collaboration meta-analysis of 40,597 trial participants with CKD recruited into antiplatelet versus placebo trials and 11,805 recruited into antiplatelet agent comparison trials found that allocation to antiplatelet therapy may reduce the RR of myocardial infarction by approximately 12% (RR: 0.88; 95% CI: 0.79–0.99). There was an expected increased risk of major bleeding,

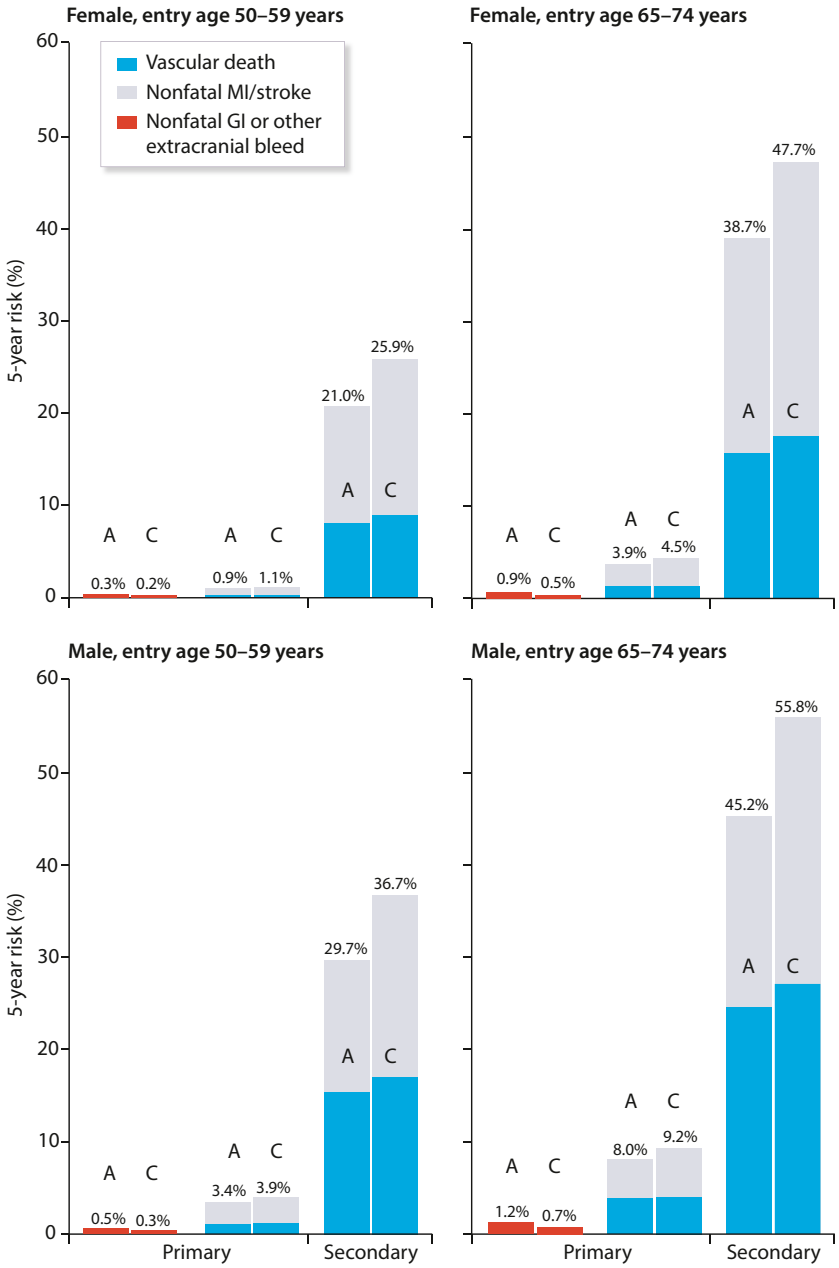

Figure 38 | Predicted 5-year absolute benefits and harms of allocation to aspirin (*A*) versus control (*C*) using a secondary or primary prevention strategy, by different levels of risk (based on age and sex). GI, gastrointestinal; MI, myocardial infarction. Reproduced with permission from *The Lancet*, volume 373, Antithrombotic Trialists' (ATT) Collaboration, Aspirin in the primary and secondary prevention of vascular disease: collaborative meta-analysis of individual participant data from randomised trials, pages 1849–1860, Copyright © 2009 Elsevier Ltd.[685a]

but the magnitude of the RR was consistent with the data from general populations (RR: 1.35; 95% CI: 1.10–1.65).[686] Note that these analyses did not distinguish between primary and secondary prevention settings.[686] The 2009 Anti-thrombotic Treatment Trialists' collaboration meta-analysis and results from 3 more recent large trials (A Study of Cardiovascular Events in Diabetes [ASCEND],[687] Aspirin in Reducing Events in the Elderly [ASPREE],[688] and Aspirin to Reduce Risk of Initial Vascular Events [ARRIVE][689]) assessing the effects of aspirin versus placebo for primary prevention in specific high-risk populations found that any harm from major bleeding counterbalanced any benefit of aspirin on cardiovascular risk (with ASPREE and ARRIVE both finding no significant effect on cardiovascular events in their studied populations of older adults or high-risk adults, respectively).[685a] A dedicated large primary prevention aspirin trial in CKD is underway.[690]

Certainty of evidence. The 2009 meta-analysis by the Anti-thrombotic Treatment Trialists' collaboration on the effect of aspirin compared with placebo in terms of the primary and secondary prevention of CVD and safety among people with and without CKD was assessed to have high risk of bias using the Risk of Bias Assessment Tool for Systematic Reviews (ROBIS) checklist due to unclear identification and selection of studies, unclear data collection and study appraisal, and high risk of bias for synthesis and findings (although we did not contact the authors to clarify these details).[685a] This review did not report on the evidence or certainty of evidence assessments directly in the report. Given the available evidence, the recommendation has a low certainty of evidence (Level C).

Values and preferences. Maintaining QoL by minimizing risk of worsening of ischemic heart disease and recurrent stroke-related disability is important to both people with CKD and caregivers.[691] The Work Group considered that the risk of bleeding would be considered acceptable by most people with CKD once the clear net benefits were explained and gastroprotection was offered. The Work Group considered that some people with CKD without prior ischemic coronary, cerebrovascular, or peripheral arterial disease but at increased risk (e.g., due to diabetes) may still wish to consider using aspirin and accept the risk of major bleeding.[687] Some people with CKD may also have a kidney diagnosis that indirectly supports considering the use of aspirin despite a lack of evidence (e.g., presumed or proven renovascular disease). The Work Group is not aware of any risk tools that could be used to help counsel such people with CKD as to their expected net absolute benefits and risks based on risk factors of the person with CKD, including any difference by sex. (Note that scores to predict cardiovascular risk are considered in Chapter 2.)

Resource use and costs. Low-dose aspirin is available at low cost and does not require monitoring.

Considerations for implementation. Proton-pump inhibitors (PPIs) are generally effective,[692] safe, and low cost (although occasionally associated with an interstitial nephritis), and the Work Group considers that it is prudent to consider bleeding risk and offers PPIs when prescribing antiplatelet therapy or antithrombotic therapy, particularly when such therapies are combined.[693]

Rationale

Meta-analysis of trials has clearly established the cardiovascular benefits of low-dose aspirin in people who have established ASCVD. Any harm of bleeding is far outweighed by the benefits (unlike the situation for primary prevention, where bleeding risk has been consistently identified in large aspirin trials and cardiovascular benefits to date have not).

Practice Point 3.15.2.1: Consider other antiplatelet therapy (e.g., $P2Y_{12}$ inhibitors) when there is aspirin intolerance.

Bleeding from gastrointestinal mucosa with antiplatelet therapy is likely to be due to their effect on hemostasis of pre-existing mucosal lesions. This hypothesis is supported by $P2Y_{12}$ inhibitors (e.g., clopidogrel or ticagrelor) not reducing the risk of bleeding in trials comparing them to aspirin.[694,695] However, if people are aspirin intolerant, a $P2Y_{12}$ inhibitor is a noninferior alternative. Note that in 2009, the FDA recommended that the coadministration of clopidogrel and omeprazole (a PPI) should be avoided because omeprazole reduces the effectiveness of clopidogrel. There is uncertainty about the precise effect of omeprazole as pharmacokinetic data are inconclusive, but PPIs with inhibition of CYP2C19 are preferred when using clopidogrel.[696]

Guidelines from the cardiology community provide recommendations for the use of dual antiplatelet therapy for a period after acute coronary syndrome or percutaneous coronary intervention. These guidelines recommend to apply the same diagnostic and therapeutic strategies in people with CKD.[663] CKD does not modify the benefits of ticagrelor,[697] and antiplatelet therapy doses do not need to be modified at decreased eGFR. Note that other antithrombotic therapy choices and doses may need to consider a person's GFR.

Special considerations

International considerations. Given the clinical effectiveness of low-dose aspirin and its low cost, there should not be many barriers to accessing this medication in any setting.

3.15.3 Invasive versus intensive medical therapy for coronary artery disease

Recommendation 3.15.3.1: We suggest that in stable stress-test confirmed ischemic heart disease, an initial conservative approach using intensive medical therapy is an appropriate alternative to an initial invasive strategy *(2D)*.

This recommendation places high value on the finding from recent, large trials in both general and CKD populations that have suggested that intensive medical therapy is a suitable initial strategy for the management of stable stress-test confirmed ischemic heart disease. It places value on the need for interventions, which carry risk to people with CKD and substantial healthcare costs to demonstrate benefits on cardiovascular outcomes before they are considered a standard of care. Importantly, this recommendation should not apply to those with severe angina symptoms, left ventricular dysfunction (e.g., ejection fraction <35%), or left main stem disease as they were excluded from the definitive trials. It should be noted that trials in CKD have not ruled out antianginal benefits in people with CKD (despite negative findings).

Key information

Balance of benefits and harm. *Benefits.* Benefits should be considered in the context of the totality of evidence in people with and without CKD regarding interventions. Comparisons between aggressive medical therapy alone and invasive interventions do not support invasive strategies to reduce death or prevent myocardial infarction.[707,707a] However, those with frequent angina symptoms (at least weekly) gained improvement with the invasive strategy[707]; thus, the benefit of an invasive strategy might be restricted to those with angina. The reason for a lack of clear antianginal effect of an invasive strategy in International Study of Comparative Health Effectiveness with Medical and Invasive Approaches—Chronic Kidney Disease (ISCHEMIA-CKD) needs some consideration, and key reasons relating to insufficient power due to protocol differences have been proposed.[698] Although low power to detect an effect on angina is a key potential explanation for differences in findings between the 2 trials, CKD-MBD and coronary calcification in CKD, which makes microvascular disease more common and increases the technical challenge of revascularization, may also have partly contributed to these differences.[699]

The ERT assessed the effects of angiography or coronary intervention in people with CKD and ischemic heart disease identified 4 other trials, but excluded mixed populations, including ISCHEMIA-CKD, which recruited some people on dialysis and some people who have received a kidney transplant. The review found no clear benefits on cardiovascular outcomes in 3 other trials and raised a hypothesis about beneficial effects on mortality overall (Supplementary Table S13[700–704]). Such an effect has not been observed in the larger general population trials.

Harms. The harms of invasive strategies include the risk of dialysis initiation, death, and stroke risk (stroke was interestingly not periprocedural).[707]

Certainty of evidence. The ERT review was limited to trials only recruiting people with CKD (and did not include the ISCHEMIA-CKD trial discussed above due to the inclusion of some people on dialysis and some people who have received a kidney transplant). The overall certainty of the evidence comparing coronary revascularization with optimal medical therapy among people with CKD not undergoing KRT and ischemic heart disease is very low (Supplementary Table S13[700–704]). Most of the RCTs reporting on the critical outcomes (all-cause mortality, CVD mortality, CVD events, kidney failure, and AKI) had some concerns regarding the risk of bias, particularly with lack of blinding for the outcome assessors, participants crossing over to the other treatment group, and the selection of reporting. The certainty of the evidence was downgraded for all outcomes because of imprecision. The certainty of the evidence for cardiovascular mortality was downgraded because publication bias was strongly suspected.

Values and preferences. Although this was not confirmed by ISCHEMIA-CKD, antianginal benefits of an invasive strategy are apparent in general populations, and people with symptoms may still elect for an initially invasive approach to manage stable stress-test confirmed coronary artery disease after being counseled about the risks.

Resource use and costs. It is not possible to formally assess the cost-effectiveness of intensive medical therapy versus an initial invasive strategy due to mixed findings from the evidence in people with stable ischemic heart disease. However, invasive strategies will have higher cost implications to healthcare systems, people with CKD, or both.

Considerations for implementation. Access and availability of invasive therapies will vary in different healthcare systems, as might the availability of medications for maximal medical therapy. The key to implementation is to encourage the understanding of the value of full therapy as compared with invasive therapy so that healthcare providers and people with CKD understand the risks and benefits of invasive strategies. Given the costs of invasive strategies, there may be additional value to implementing this recommendation.

Rationale

Evidence suggests that the key indication for an initial invasive strategy to manage stable ischemic heart disease is based on symptoms, and intensive medical therapy is a suitable approach if symptom control is satisfactory in people with or without CKD. In CKD, the antianginal benefits of an initially invasive approach have not been demonstrated.

Practice Point 3.15.3.1: Initial management with an invasive strategy may still be preferable for people with CKD with acute or unstable coronary disease, unacceptable levels of angina (e.g., patient dissatisfaction), left ventricular systolic dysfunction attributable to ischemia, or left main disease.

The ISCHEMIA trial has been described as deeply disrupting prior attitudes regarding management strategies for people with stable coronary artery disease,[705] and clinical practice guidelines that predate the trial need updating.[706] Despite the International Study of Comparative Health Effectiveness with Medical and Invasive Approaches (ISCHEMIA) and ISCHEMIA-CKD trial results, it is

considered that the well-established intervention of coronary revascularization will continue to have a key role in angina relief.[705] Importantly, this recommendation should not apply to those with unacceptably severe angina symptoms. It should also be noted that people with left ventricular dysfunction (i.e., ejection fraction <35%) or left main disease were excluded from the definitive ISCHEMIA trial.[707a] The Work Group considers that certain design features of the ISCHEMIA-CKD trial may have led to angina benefits not being detected, and the trial results should not rule out angina benefits in people with CKD (see above). If an invasive strategy is pursued, there are effective strategies to reduce the risk of contrast-induced AKI (Chapter 4).[708]

The totality of the evidence from the CKD-specific trials is consistent with no net difference between an initial conservative approach using aggressive medical therapy versus an invasive strategy when treating stable stress-test confirmed ischemic heart disease. This is consistent with the large general population-based ISCHEMIA trial.[707a]

Overall eGFRcr	ACR (mg/g) <10	10–29	30–299	300–999	1000+
105+	1.1	1.3	1.7	2.4	3.5
90–104	Ref	1.2	1.5	1.9	2.3
60–89	1.0	1.2	1.4	1.7	2.2
45–59	1.2	1.3	1.5	1.8	2.4
30–44	1.4	1.5	1.7	2.0	2.4
15–29	1.9	1.8	2.0	2.6	3.0
<15	2.6	2.5	3.1	3.6	4.2

Atrial fibrillation: 50 cohorts
Study size = 22,886,642; events = 1,068,701

Figure 39 | Meta-analyzed adjusted prevalence of atrial fibrillation from cohorts contributing to the Chronic Kidney Disease Prognosis Consortium, by diabetes status. ACR, albumin-to-creatine ratio eGFRcr, creatinine-based estimate glomerular filtration rate. Reproduced with permission from *JAMA*, Writing Group for the CKD Prognosis Consortium; Grams ME, Coresh J, Matsushita K, et al. Estimated glomerular filtration rate, albuminuria, and adverse outcomes: an individual-participant data meta-analysis. *JAMA*. 2023;330(13):1266–1277.[12] Copyright © 2023 American Medical Association. All rights reserved.

3.16 CKD and atrial fibrillation

In CKD, the same principles to diagnose and manage atrial fibrillation should be used as in people without CKD.

Prevalence and consequences. Atrial fibrillation is the commonest sustained arrhythmia, with risk increasing steeply with increasing age (earlier in men than women).[709] There is a particularly high prevalence in people with CKD. Crude prevalence ranging from 16% to 21% has been reported in people with CKD not requiring KRT.[710] In the cohorts contributing to the CKD-PC, adults with CKD G3, A1 had an adjusted risk of atrial fibrillation of 1.2–1.5, increasing to an adjusted risk of 4.2 by CKD stages G5, A3 (Figure 39[12]). Atrial fibrillation can directly cause thromboembolism (particularly stroke) and/or heart failure. It is also linked, perhaps directly or through shared risk factors, with increased risk of death, hospitalization, vascular dementia, depression, and reduced QoL.[709] Detailed clinical practice guidelines have been formulated by the cardiology community describing definitions, classification, diagnosis, screening strategies, and management.[709] It is beyond the scope of this KDIGO guideline to consider all aspects of the diagnosis and management of atrial fibrillation in people with CKD. The ERT review focused on the role of non–vitamin K antagonist oral anticoagulants (NOACs) versus warfarin for thromboprophylaxis in CKD.

Identification and management. Atrial fibrillation can be asymptomatic but symptoms are not a prerequisite for risk of complications. As the prevalence of atrial fibrillation is high in people with CKD and there are effective strategies to manage its associated complications, opportunistic pulse-based screening (e.g., when taking BP), followed by a 12-lead electrocardiogram if an irregularly irregular pulse is identified should be considered. Such an approach is low cost and simple to implement. Figure 40 outlines approaches to different diagnostic and management strategies.

Practice Point 3.16.1: Follow established strategies for the diagnosis and management of atrial fibrillation (Figure 40).

Prophylaxis against stroke and systemic thromboembolism. Recent cardiology guidelines recommend a risk factor–based approach to stroke thromboprophylaxis decisions in atrial fibrillation using the Congestive heart failure, Hypertension, Age ≥75 (doubled), Diabetes, Stroke (doubled), Vascular disease, Age 65 to 74, and Sex category (female) (CHA_2DS_2-VASc) stroke risk score. They recommend that only people at "low stroke risk" (CHA_2DS_2-VASc score = 0 in men, or 1 in women) should not be offered antithrombotic therapy. Oral anticoagulants should be considered for stroke prevention with a CHA_2DS_2-VASc score of 1 in men or 2 in women, considering net clinical benefit and values and preferences of people with CKD. Oral anticoagulants are clearly recommended for stroke prevention in people with atrial fibrillation and a CHA_2DS_2-VASc score ≥2 in men or ≥3 in women.[709] Our Work Group considered that oral anticoagulation for thromboprophylaxis should nearly always be considered for preventing stroke in people with decreased eGFR and atrial fibrillation (Figure 40). The presence of decreased GFR is a risk for thromboembolic stroke in people with atrial fibrillation.[710–712] It has been estimated that approximately 95% of people with an eGFR of <60 ml/min per 1.73 m^2 have a CHA_2DS_2-VASc score of ≥2, increasing to approximately 99% at an eGFR of <30 ml/min per 1.73 m^2.[711] Importantly, it has also been shown that in a group of people with a CHA_2DS_2-VASc score of 0 to 1 point (i.e., a group where thromboprophylaxis may not be considered indicated), people with CKD within the group are at much higher risk of cerebrovascular and other systemic

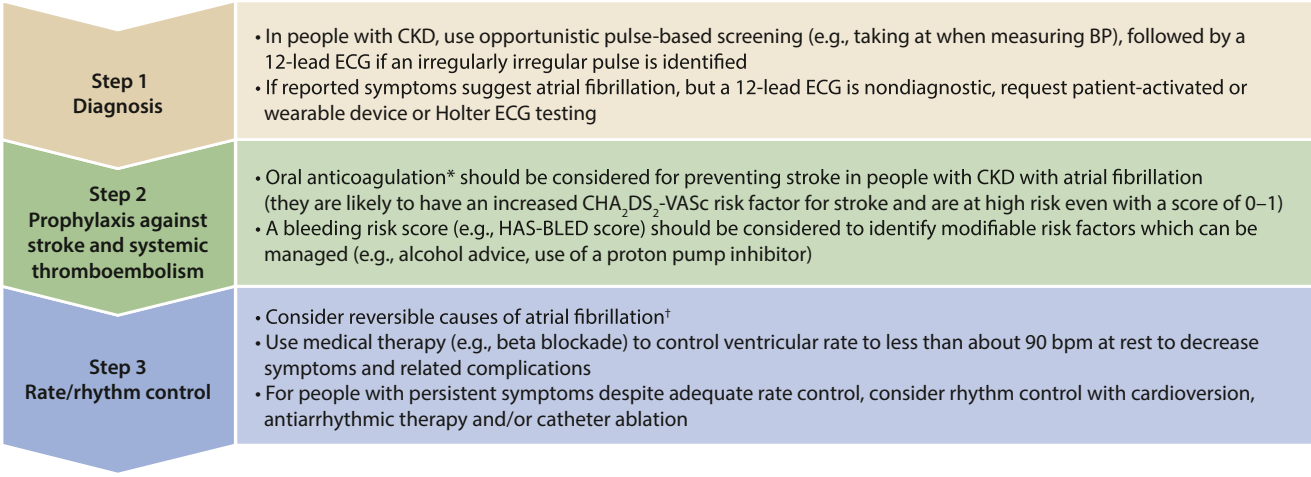

Figure 40 | Strategies for the diagnosis and management of atrial fibrillation. *Consider dose adjustments necessary in people with chronic kidney disease (CKD). †The following has been recommended as a standard package for diagnostic evaluation of new atrial fibrillation: (i) a 12-lead electrocardiogram (ECG) to establish the diagnosis, assess ventricular rate, and check for the presence of conduction defects, ischemia, or structural heart disease; (ii) laboratory testing for thyroid and kidney function, serum electrolytes, and full blood count; and (iii) transthoracic echocardiography to assess left ventricular size and function, left atrial size, for valvular disease, and right heart size and function. BP, blood pressure; CHA_2DS_2-VASc, Congestive heart failure, Hypertension, Age ≥75 (doubled), Diabetes, Stroke (doubled), Vascular disease, Age 65 to 74, and Sex category (female); HAS-BLED, Hypertension, Abnormal liver/kidney function, Stroke history, Bleeding history or predisposition, Labile international normalized ratio (INR), Elderly, Drug/alcohol usage.

thromboembolic events, with an annual rate of 2.9% compared with 0.2% in people without CKD.[711]

Including GFR into atrial fibrillation risk scores has not shown important incremental benefit to its introduction (e.g., adding 2 points for CrCl <60 ml/min to $CHADS_2$—referred to as Renal Dysfunction, Congestive Heart Failure, Hypertension, Age, Diabetes, Stroke/Transient Ischemic Attack [R_2CHADS_2])—improved net reclassification index but not the C-statistic.[710] However, as decreased GFR is associated with age, diabetes, CVD, and so on, the incremental predictive advantage by adding a CKD parameter to the CHA_2DS_2-VASc score, which already includes these parameters, would be expected to have little effect. There is considerable scope to improve the predictive performance of thromboprophylaxis risk scores for use in CKD.[713]

Recommendation 3.16.1: We recommend use of non–vitamin K antagonist oral anticoagulants (NOACs) in preference to vitamin K antagonists (e.g., warfarin) for thromboprophylaxis in atrial fibrillation in people with CKD G1–G4 *(1C)*.

This recommendation puts high value on the use of NOACs, also referred to as direct-acting oral anticoagulants or DOACs, in people with CKD due to their simpler pharmacokinetic profile, dosing, and monitoring than vitamin K antagonists and due to their improved efficacy and relatively similar safety profile. Although people with CKD G4–G5 have been understudied in RCTs, implementation in such groups can be achieved after considering choice of NOAC and dosing.

Key information

Balance of benefits and harms. *Benefits.* Data from 42,411 participants who received NOACs and 29,272 participants who received warfarin in 4 phase III trials were meta-analyzed in 2014. Such trials largely excluded people with CKD G4–G5 but did include large numbers of participants with earlier stages of CKD. Overall, NOACs significantly reduced the risk of stroke or systemic embolic events by 19% compared with warfarin (RR: 0.81; 95% CI: 0.73–0.91). This benefit was a result largely from reduced risk of hemorrhagic strokes (RR: 0.49; 95% CI: 0.38–0.64). There were large amounts of data on stroke in those with a CrCl of <50 ml/min, and the relative benefits were consistent and clearly evident in people with CKD. There were also consistent effects in subgroup analyses by age, sex, prior diabetes, prior stroke, and $CHADS_2$ score.[714] A more recent meta-analysis published in 2021 only focused on subgroups with CKD and included data from 7 trials of NOACs versus warfarin in atrial fibrillation. It also reported a 19% reduced risk of stroke/thromboembolic complications in the NOAC group (HR: 0.81; 95% CI: 0.69–0.97).[715] Data in CKD G5 on dialysis were limited to observational studies.[715] Our evidence review aimed to collect information on subtypes of outcome from subgroup analyses reporting results specifically in people with CKD. Evidence of efficacy in the large trials is mainly for the outcomes of stroke and hemorrhagic stroke, but our review only found data from 3 trials for these outcomes resulting in imprecise estimates of effect. The findings were qualitatively consistent with the totality of the evidence (Figure 41, Supplementary Table S14[716–721]).

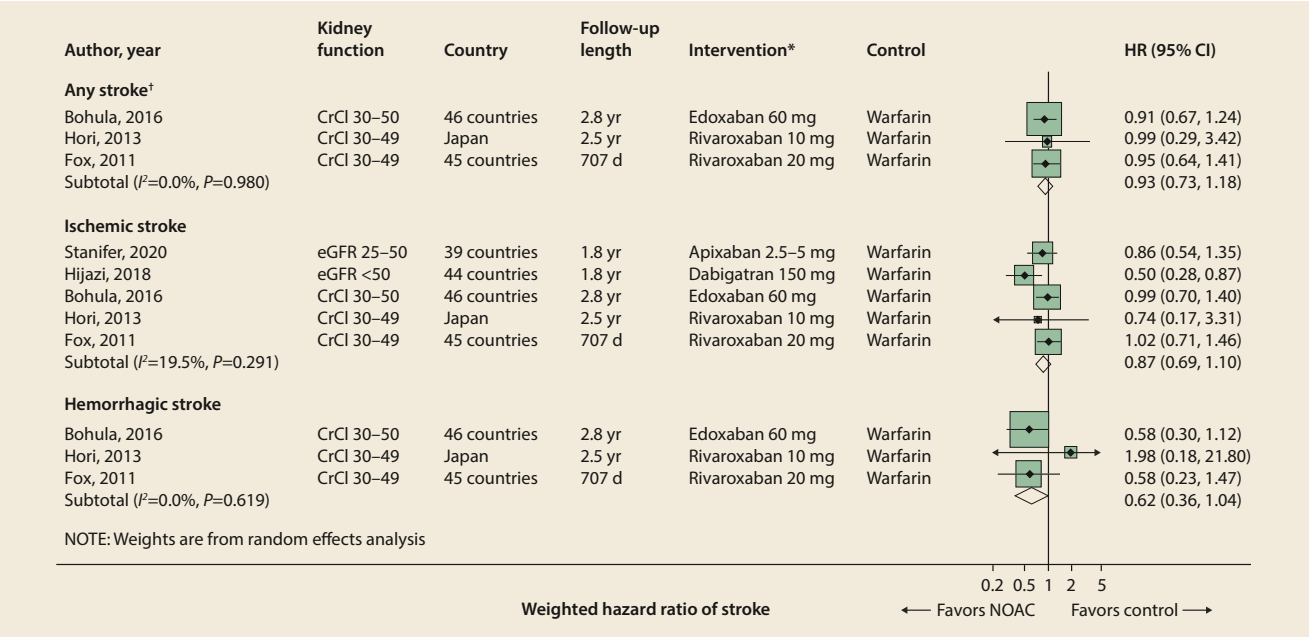

Figure 41 | Pooled hazard ratio (HR) comparing non–vitamin K antagonist oral anticoagulants (NOACs) with warfarin among people with chronic kidney disease in terms of stroke. Bohula E, Giugliano R, Ruff C, et al. Impact of renal function on outcomes with edoxaban in the ENGAGE AF-TIMI 48 trial. *Circulation*. 2016;134:24–36[716]; Fox KA, Piccini JP, Wojdyla D, et al. Prevention of stroke and systemic embolism with rivaroxaban compared with warfarin in patients with non-valvular atrial fibrillation and moderate renal impairment. *Eur Heart J*. 2011;32:2387–2394[718]; Hijazi Z, Hohnloser SH, Oldgren J, et al. Efficacy and safety of dabigatran compared with warfarin in patients with atrial fibrillation in relation to renal function over time—a RE-LY trial analysis. *Am Heart J*. 2018;198:169–177[719]; Hijazi Z, Alexander JH, Li Z, et al. Apixaban or vitamin K antagonists and aspirin or placebo according to kidney function in patients with atrial fibrillation after acute coronary syndrome or percutaneous coronary intervention: insights from the AUGUSTUS trial. *Circulation*. 2021;143:1215–1223[722]; Hori M, Matsumoto M, Tanahashi N, et al. Safety and efficacy of adjusted dose of rivaroxaban in Japanese patients with non-valvular atrial fibrillation: subanalysis of J-ROCKET AF for patients with moderate renal impairment. *Circ J*. 2013;77:632–638[720]; Stanifer J, Pokorney S, Chertow G, et al. Apixaban versus warfarin in patients with atrial fibrillation and advanced chronic kidney disease. *Circulation*. 2020;141:1384–1392.[721] CI, confidence interval; CrCl, creatinine clearance; eGFR, estimated glomerular filtration rate.

Harms. The 2014 meta-analysis of 4 large phase III trials found that NOACs reduced the risk of death from any cause by 10%, confirming net safety (RR: 0.90; 95% CI: 0.85–0.95). Compared with warfarin, NOACs reduced the risk of intracranial hemorrhage (defined as hemorrhagic stroke, epidural, subdural, and subarachnoid hemorrhage) by about one-half (RR: 0.48; 95% CI: 0.39–0.59), and the risk of gastrointestinal bleeding was increased by about one-quarter (RR: 1.25; 95% CI: 1.01–1.55). Overall, there was no clear effect on the combination of these 2 safety outcomes referred to as major bleeding (RR: 0.86; 95% CI: 0.73–1.00).[714] There were large amounts of data on major bleeding in those with a CrCl of <50 ml/min, so reassuring safety data clearly extended to people with CKD. There were also consistent safety data in subgroup analyses by age, sex, prior diabetes, prior stroke, and $CHADS_2$ score. There was a suggestion that major bleeding was significantly reduced in people attending centers where time in therapeutic international normalized ratio (INR) range was <66% compared with centers with ≥66% time in range (interaction $P = 0.02$). This suggests that benefits of NOACs are in part a result of their simpler pharmacokinetic profile and dosing.[714] The 2021 meta-analysis that focused on CKD subgroups from 7 trials found that bleeding events were also not significantly different among those allocated NOACs versus warfarin (HR: 0.83; 95% CI: 0.58–1.18).[715] Data in CKD G5 on dialysis were limited to observational studies.[715] Our evidence review was again limited to a small number of studies reporting subtypes of bleeding outcomes, and so analyses found imprecise estimates of treatment effect. The findings were qualitatively consistent with the totality of the evidence (Figure 42, Supplementary Table S15[716–722]). The review raised a hypothesis that some NOACs may be more likely to reduce the risk of bleeding. However, given the evidence of effect modification by time in therapeutic range in the warfarin group, we have not provided specific recommendations to prefer certain NOACs.

Certainty of evidence. The overall certainty of the evidence comparing NOACs with warfarin among people with CKD and atrial fibrillation is low (Supplementary Tables S14[716–721] and S15[716–722]). Most of the RCTs evaluating the critical outcomes were considered to have a low risk of bias. The critical outcome of stroke was reported as any stroke, ischemic stroke, and/or hemorrhagic stroke. Because there

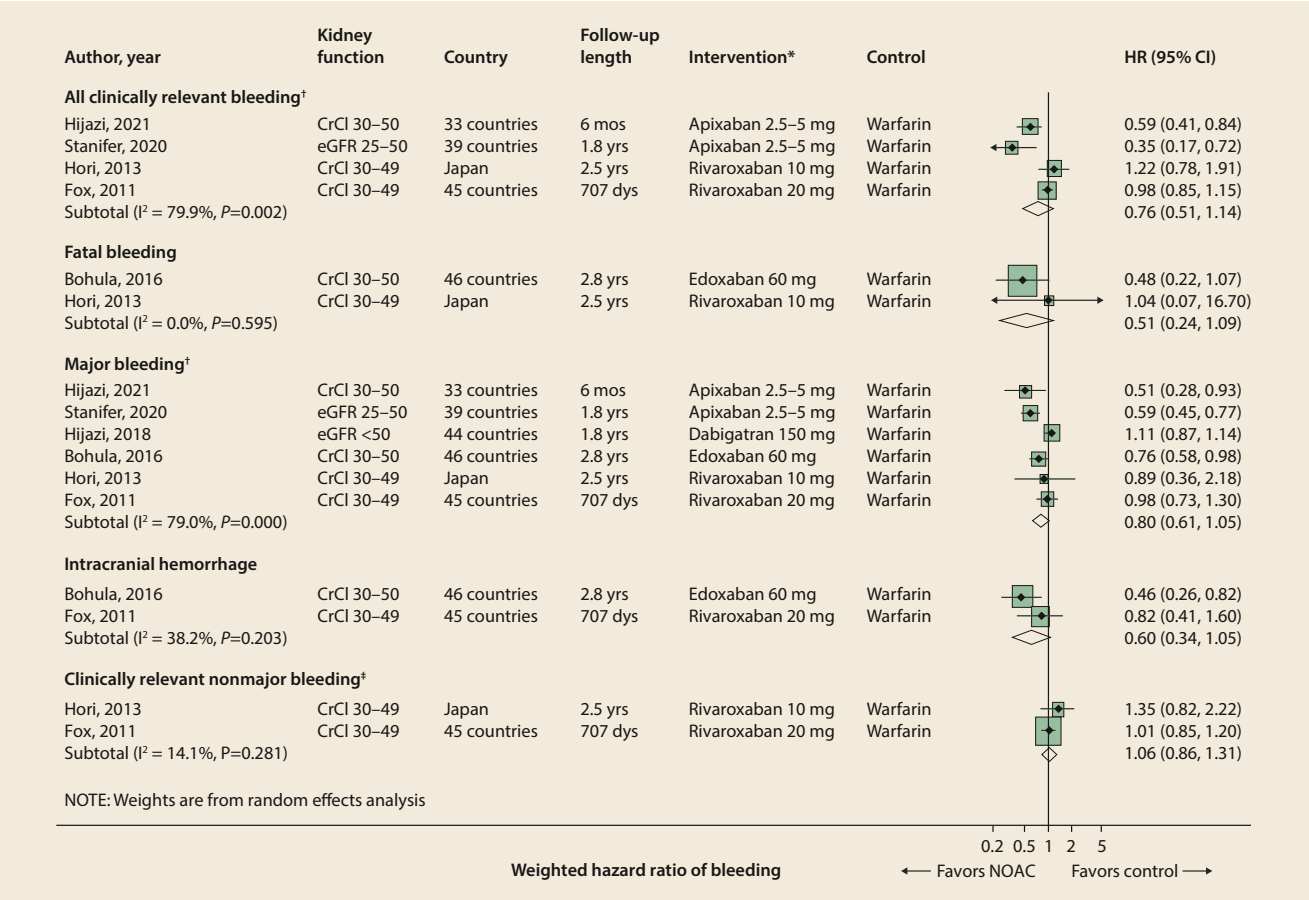

Figure 42 | Pooled hazard ratio (HR) comparing non–vitamin K antagonist oral anticoagulants (NOACs) with warfarin among people with chronic kidney disease in terms of bleeding. Bohula E, Giugliano R, Ruff C, et al. Impact of renal function on outcomes with edoxaban in the ENGAGE AF-TIMI 48 trial. *Circulation*. 2016;134:24–36[716]; Fox KA, Piccini JP, Wojdyla D, et al. Prevention of stroke and systemic embolism with rivaroxaban compared with warfarin in patients with non-valvular atrial fibrillation and moderate renal impairment. *Eur Heart J*. 2011;32:2387–2394[718]; Hijazi Z, Hohnloser SH, Oldgren J, et al. Efficacy and safety of dabigatran compared with warfarin in patients with atrial fibrillation in relation to renal function over time—a RE-LY trial analysis. *Am Heart J*. 2018;198:169–177[719]; Hori M, Matsumoto M, Tanahashi N, et al. Safety and efficacy of adjusted dose of rivaroxaban in Japanese patients with non-valvular atrial fibrillation: subanalysis of J-ROCKET AF for patients with moderate renal impairment. *Circ J*. 2013;77:632–638[720]; Stanifer J, Pokorney S, Chertow G, et al. Apixaban versus warfarin in patients with atrial fibrillation and advanced chronic kidney disease. *Circulation.* 2020;141:1384–1392[721]; Hijazi Z, Alexander JH, Li Z, et al. Apixaban or vitamin K antagonists and aspirin or placebo according to kidney function in patients with atrial fibrillation after acute coronary syndrome or percutaneous coronary intervention insights from the AUGUSTUS trial. *Circulation.* 2021;143:1215–1223.[722] CI, confidence interval; CrCl, creatinine clearance; eGFR, estimated glomerular filtration rate.

were few stroke events reported across the RCTs, the certainty of the evidence was downgraded for imprecision.

Values and preferences. High value on the use of NOACs included the conclusion that the simple dosing and lack of INR monitoring compared with vitamin K antagonists would lead to a substantial reduction in burden for those with an indication for anticoagulation and their health services. There is also good evidence for improved efficacy and a relatively similar safety profile. Most fully informed people with CKD would be expected to select a NOAC over a vitamin K antagonist.

Resource use and costs. NOACs have been shown to be cost-effective for stroke prevention in atrial fibrillation and may even be cost-saving in people with CKD. Vitamin K antagonist use may be associated with higher costs and achieve fewer quality-adjusted life-years compared with NOACs.[723]

Considerations for implementation. A decision not to anticoagulate for thromboembolic prophylaxis due to low risk would ideally be re-evaluated at each consultation and at least every 6 months. When using antithrombotic therapy in people with CKD, it is prudent to treat modifiable risk factors for bleeding (e.g., alcohol intake) and use gastroprophylaxis with a PPI, particularly when combined with antiplatelet therapy.

Rationale

A number of large RCTs demonstrated that NOACs reduce the risk of intracranial bleeding compared with warfarin and, overall, modestly reduce mortality in people with atrial

a

eCrCl (ml/min)[a]	Warfarin	Apixaban[b]	Dabigatran	Edoxaban[c]	Rivaroxaban
>95	Adjusted dose (INR 2–3)	5 mg b.i.d.	150 mg b.i.d.	60 mg QD[d]	20 mg QD
51–95	Adjusted dose (INR 2–3)	5 mg b.i.d.	150 mg b.i.d.	60 mg QD	20 mg QD
31–50	Adjusted dose (INR 2–3)	5 mg b.i.d. (eCrCl cut off 25 ml/min)	150 mg b.i.d. or 110 mg b.i.d.[e]	30 mg QD	15 mg QD

b

eCrCl (ml/min)[a]	Warfarin	Apixaban[b]	Dabigatran	Edoxaban	Rivaroxaban
15–30	Adjusted dose for INR 2–3 could be considered	2.5 mg PO b.i.d. could be considered	Unknown (75 mg PO b.i.d.)[f,g]	30 mg QD[h] could be considered	15 mg QD could be considered
<15 not on dialysis	Equipoise based on observational data and meta-analysis	Unknown (2.5 mg PO b.i.d.)[f]	Not recommended	Not recommended	Unknown (15 mg QD)[f]
<15 on dialysis	Equipoise based on observational data and meta-analysis	Unknown (2.5 mg PO b.i.d.)[f]	Not recommended	Not recommended	Unknown (15 mg QD)[f]

Figure 43 | Evidence from (a) randomized controlled trials (RCTs) regarding therapeutic anticoagulation dose by glomerular filtration rate (GFR) and (b) in areas where RCTs are lacking. Dosing of non–vitamin K antagonist oral anticoagulants (NOACs) based solely on limited pharmacokinetic and pharmacodynamic data (i.e., there are no randomized efficacy or safety trial data assessing clinical outcomes for stroke thromboprophylaxis in atrial fibrillation at chronic kidney disease [CKD] G4–G5). [a]Cockcroft-Gault estimated creatinine clearance (eCrCl). [b]Apixaban dose modification from 5 mg twice per day (b.i.d.) to 2.5 mg b.i.d. if a person has any 2 of the following: serum creatinine ≥1.5 mg/dl (133 μmol/l), age ≥80 years, or body weight ≤60 kg. [c]In the Effective Anticoagulation With Factor Xa Next Generation in Atrial Fibrillation–Thrombolysis in Myocardial Infarction 48 (ENGAGE-AF TIMI 48) study, the dose was halved if any of the following: eCrCl of 30–50 ml/min, body weight ≤60 kg, or concomitant use of verapamil or quinidine (potent P-glycoprotein inhibitors). [d]This dose has not been approved for use by the US Food and Drug Administration (FDA) in this category of GFR. [e]In countries where 110 mg b.i.d. is approved, healthcare providers may prefer this dose after clinical assessment of thromboembolic versus bleeding risk. This dose has not been approved for use by the US FDA. [f]NOAC doses listed in parenthesis are doses that do not currently have any clinical or efficacy data. The doses of NOACs apixaban 5 mg b.i.d.,[b] rivaroxaban 15 mg every day, and dabigatran 75 mg b.i.d. are included in the US FDA–approved labeling based on limited dose pharmacokinetic and pharmacodynamics data with no clinical safety data. We suggest consideration of the lower dose of apixaban 2.5 mg oral b.i.d. in CKD G5 and G5D to reduce bleeding risk until clinical safety data are available. [g]Dabigatran 75 mg available only in the United States. [h]The dose was halved if any of the following: estimated CrCl of 30–50 ml/min, body weight of ≤60 kg, or concomitant use of verapamil or quinidine (potent P-glycoprotein inhibitors). INR, international normalized ratio; QD, every day. Reproduced from Turakhia MP, Blankestijn PJ, Carrero JJ, et al. Chronic kidney disease and arrhythmias: conclusions from a Kidney Disease: Improving Global Outcomes (KDIGO) Controversies Conference. *Eur Heart J*. 2018;39:2314–2325.[710] © The Author(s) 2018. Published by Oxford University Press on behalf of the European Society of Cardiology. This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (http://creativecommons.org/licenses/by-nc/4.0/).

fibrillation. They offer benefits in terms of ease of monitoring. CKD does not appear to importantly modify these benefits, at least down to G4.

Practice Point 3.16.2: NOAC dose adjustment for GFR is required, with caution needed at CKD G4–G5.

Doses of NOACs may need to be modified in people with decreased GFR taking into consideration the age, weight, and GFR of a person with CKD (Figure 43[710]). Consult the relevant summaries of product characteristics for the latest information on dosing (Chapter 4).

Practice Point 3.16.3: Duration of NOAC discontinuation before elective procedures needs to consider procedural bleeding risk, NOAC prescribed, and level of GFR (Figure 44).[710,724]

	Dabigatran		Apixaban–Edoxaban–Rivaroxaban	
	No important bleeding risk and/or adequate local hemostasis possible: perform at trough level (i.e., ≥12 or 24 h after last intake)			
	Low risk	High risk	Low risk	High risk
CrCl ≥80 ml/min	≥24 h	≥48 h	≥24 h	≥48 h
CrCl 50–80 ml/min	**≥36 h**	**≥72 h**	≥24 h	≥48 h
CrCl 30–50 ml/min[a]	**≥48 h**	**≥96 h**	≥24 h	≥48 h
CrCl 15–30 ml/min[a]	No official indication	No official indication	**≥36 h**	**≥48 h**
CrCl <15 ml/min	No official indication for use There is no need for bridging with LMWH/UFH			

Figure 44 | Advice on when to discontinue non–vitamin K antagonist oral anticoagulants (NOACs) before procedures (low vs. high risk). The bold values deviate from the common stopping rule of ≥24-hour low risk, ≥48-hour high risk. Low risk is defined as a low frequency of bleeding and/or minor impact of a bleed. High risk is defined as a high frequency of bleeding and/or important clinical impact. Adapted from Heidbuchel H, Verhamme P, Alings M, et al. Updated European Heart Rhythm Association practical guide on the use of non–vitamin-K antagonist anticoagulants in patients with non-valvular atrial fibrillation: executive summary. *Eur Heart J.* 2017;38:2137–2149.[724] [a]Many of these people may be on lower dose of dabigatran (110 mg twice per day [b.i.d.]) or apixaban (2.5 mg b.i.d.), or have to be on the lower dose of rivaroxaban (15 mg QD) or edoxaban (30 mg QD). Dabigatran 110 mg b.i.d. has not been approved for use by the US Food and Drug Administration. CrCl, creatinine clearance, LMWH, low-molecular-weight heparin; UFH, unfractionated heparin. Reproduced from Turakhia MP, Blankestijn PJ, Carrero JJ, et al. Chronic kidney disease and arrhythmias: conclusions from a Kidney Disease: Improving Global Outcomes (KDIGO) Controversies Conference. *Eur Heart J.* 2018;39:2314–2325.[710] © The Author(s) 2018. Published by Oxford University Press on behalf of the European Society of Cardiology. This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (http://creativecommons.org/licenses/by-nc/4.0/).

For research recommendations, please see Chapter 6: Research recommendations.